OM DWIVEDI

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PROFILE SUMMARY

Final-year B.Tech Computer Science Engineering (AI & ML) graduate with hands-on experience building full-stack web applications and LLM-integrated systems using Python, Flask, SQL, and REST APIs. Co-author of a published IEEE-format research paper on an AI-powered fitness platform. Strong foundation in OOP, DBMS, and database design, with a track record of independently designing, building, and deploying end-to-end projects. Seeking a Software Development, AI/ML, or Technical Support role to deliver reliable, production-grade solutions.

TECHNICAL SKILLS

- **Programming Languages:** Python, SQL, C (basics)
- **Web Development:** Flask (REST APIs), HTML5, CSS3, Bootstrap 5, Jinja2
- **Databases:** MySQL (DDL/DML, Joins, Aggregation, Indexing), SQLite, SQLAlchemy ORM
- **AI/ML:** LLM Prompting & Integration, Groq API (LLaMA-based inference), Prompt Engineering
- **Tools & Platforms:** Git, GitHub, VS Code, Jupyter Notebook, Google Colab, Postman
- **Core CS Fundamentals:** Object-Oriented Programming, DBMS, Operating Systems, Computer Networks

EDUCATION

B.Tech, Computer Science & Engineering (AI & ML) 2022 – 2026
Babu Banarasi Das National Institute of Technology and Management, Lucknow | CGPA: 6.6
Senior Secondary (Class XII), CBSE — Swarnim Public School: 68%
Secondary (Class X), CBSE — Swarnim Public School: 84%

PROJECTS

AI-Based Personalised Fitness Recommendation System | *Python, Flask, Groq API, SQLAlchemy, MySQL, Bootstrap 5*

- Built a full-stack AI gym management web application generating personalised workout and diet plans via LLM inference using the Groq API (LLaMA-based model).
- Developed "Coach Alex", an AI coaching assistant delivering contextual fitness guidance, session-level progress tracking, and exercise form correction through prompt-engineered LLM responses.
- Implemented a gamified reward system (workout streaks to achievement badges) and a BMI-driven recommendation engine tailored to fitness goals and dietary preferences.
- Managed data using SQLAlchemy ORM with MySQL and visualised user progress with Chart.js; co-authored an IEEE-format research paper published in Metszet Journal (ISSN: 2061-2710, Vol. 11, Issue 4, 2026).

Indian Diet Calorie Tracker | *Python, Flask, Pandas, FuzzyWuzzy, SQLite, Bootstrap 5*

- Built a calorie estimation web app addressing the gap in existing apps that fail to recognise Indian regional food items such as dal tadka, poha, samosa, and regional thalis.
- Compiled a custom nutrition dataset of 500+ items from IFCT/NIN data and implemented fuzzy string matching to handle Hindi transliterations and regional name variants.
- Designed a portion-size conversion layer (katori, roti count, glass) that auto-converts to grams; built a Flask REST API with SQLite backend, daily macro breakdowns, and Chart.js weekly trend graphs.

Student & Employee Record Management System | *Python, MySQL, mysql-connector-python*

- Designed console-based CRUD applications to manage student and employee records using SQL joins, aggregate queries, and role-based data filtering.
- Applied input validation, exception handling, and parameterised queries to prevent SQL injection, using a modular code structure for maintainability.

ACHIEVEMENTS & EXTRA-CURRICULARS

- Co-authored research paper "AI-Based Personalised Fitness Recommendation System Using LLMs and Groq API", published in Metszet Journal (IEEE format, 2026).
- Solved 50+ DSA problems on LeetCode and HackerRank covering Arrays, Strings, SQL, and Sorting.
- Presented at the departmental project expo; received positive faculty feedback on the AI fitness system and database projects.
- Active GitHub contributor, maintaining public repositories for all academic and personal projects.